

SELECT with various clause – GROUP BY, HAVING, ORDER BY

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MySQL 8.0 Command Line Cli
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```
mysql> CREATE DATABASE GroupDemo;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> USE GroupDemo;
Database changed
mysql>
mysql> CREATE TABLE Student (
  -> ID INT,
  -> Name VARCHAR(30),
  -> Department VARCHAR(20),
  -> Marks INT
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO Student VALUES
  -> (1, 'Arun', 'CSE', 80),
  -> (2, 'Bala', 'CSE', 70),
  -> (3, 'Charan', 'ECE', 85),
  -> (4, 'Divya', 'ECE', 75),
  -> (5, 'Kiran', 'CSE', 90);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT * FROM Student;
+----+-----+-----+-----+
| ID | Name  | Department | Marks |
+----+-----+-----+-----+
| 1  | Arun  | CSE       | 80    |
| 2  | Bala  | CSE       | 70    |
| 3  | Charan | ECE       | 85    |
| 4  | Divya | ECE       | 75    |
| 5  | Kiran | CSE       | 90    |
+----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT Department, COUNT(*) AS Total_Students
  -> FROM Student
  -> GROUP BY Department;
+-----+-----+
| Department | Total_Students |
+-----+-----+
| CSE       | 3              |
| ECE       | 2              |
+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT Department, AVG(Marks) AS Average_Marks
  -> FROM Student
  -> GROUP BY Department
  -> HAVING AVG(Marks) > 75;
+-----+-----+
| Department | Average_Marks |
+-----+-----+
| CSE       | 80.0000       |
| ECE       | 80.0000       |
+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Student
  -> ORDER BY Marks DESC;
+----+-----+-----+-----+
| ID | Name  | Department | Marks |
+----+-----+-----+-----+
| 5  | Kiran | CSE       | 90    |
| 3  | Charan | ECE       | 85    |
| 1  | Arun  | CSE       | 80    |
| 4  | Divya | ECE       | 75    |
| 2  | Bala  | CSE       | 70    |
+----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```